SimSE Educational Game B:
Early Sizing of Requirements
(Version: March 30, 2026)

Audience: Students

Editors:

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Sponsors:

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1. Introduction/Background

This SimSE game simulate a set of measurement activities early in a software lifecycle, and game player role is that of a manager responsible of managing this set of measurement activities. This simulation is designed to evaluate the impact of the player's decisions on selecting the measures, the set requirements documents and related measurement tools selection. This simulation trains students on how to make management choices taking into consideration several factors concurrently.

2. How to Use This Module

This module is designed to be used as part of a course on software functional size measurement (FSM) and its usage in software estimation.

SimSE is intended to be used as a *complementary* component to a course, not as a standalone instructional tool.

Therefore, software functional size measurement (FSM) in general should be introduced to students either before, or in parallel with the students' exposure to the game (either through lectures (see Section 6), readings (see Section 7), or some other method).

Before students are given the assignment to play a SimSE game, it is imperative that they watch the SimSE Gameplay video tutorial. It is strongly recommended that they also watch the Explanatory Tool and Game Branching video tutorials as well. All video tutorials are available at http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Videos.

Our experience with SimSE has shown again and again how crucial the instructions a student receives in learning to play the game are to their success in learning from it.

These video tutorials have been designed to specifically highlight and address aspects of SimSE that are critical for students to understand for a maximally effective educational experience. Therefore, we suggest that you not only assign the students to watch these videos on their own time, but, if time and resources allow, show them in class as well, emphasize how important they are to watch, and also point them to the SimSE player's manual, available at http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Docs.

If time and resources further warrant, students should be required to attend a teaching-assisted (TA) -lead training session, in which they are shown the videos, given a printed player's manual and, and then allowed to try playing the game for a while with the TA (who should have already studied the manual and played SimSE themselves) available to answer any questions they may have.

Students should be given the questions to answer for this module (see Section 8) at the time they are asked to play the game. Having the questions to refer to while they play helps point them to some of the

more subtle lessons encoded in the game, as well as provides you, as an instructor, with a way to assess whether or not they have completed the assignment and learned the concepts.

One could use this module as a mandatory part of a course, or else make it an extra-credit assignment. Given that the exercise is actually quite involved, especially when multiple games are assigned, somewhere in the order of 5-10% extra credit is recommended. Certain prerequisites for this particular module apply (see Section 4).

3. Learning Objectives

These lessons were taken from software management from the software size measurement perspective. The following three game-specific lessons were implemented in this SimSE game:

- *The manager (e.g., the game player) needs to make decisions on a number of factors.*
    - Selection of two measurers as the measurement team.
        - All candidate measurers have different profiles, expertise and salaries.
            - The profiles influence their accuracy in size measurement;
            - The expertise influences their velocity in size measurement;
            - Their salaries are related to their expertise and impact the measurement costs.
    - Select how many functional user requirements (FUR) will be measured – leading to distinct budget according to the number of requirements
    - Select whether a tool will be used by the measurers: the usage of a tool will influence the speed also the cost of measurement (e.g., the budget). In this SimSE game, the tool corresponds to the document: COSMIC Early Sizing Guidelines.
    - If the requirements document selected by the game player is only at a high level of documentation - without functional details - then it is not possible to use the detailed COSMIC rules, and this specific COSMIC 'tool' is not useful in such a context, and it as no impact on the measurement budget.
        - If the player chooses to use the tool, the two measurers selected could increase their measurement speed at some associated costs.
        - However, if the player chooses to measure the requirements with only a high documentation level, COSMIC detailed rules will not be necessary, since there is not sufficient detailed requirement for measurers to perform activity - Therefore, the player would rather choose "No tool" to be more reasonable.
- Among the factors of accuracy, speed, expertise and budget, the player needs to decide through several factors.

- *How much budget left is not evaluated in the game.* The SimSE system game will assign to the measurers team (e.g., manager) an initial budget based on the number of requirements they selected to size. And both the team members' salaries and tool selection will cost money.

- The game player-manager must focus on completing the measurement before spending all of the budget.

- As long as there is still some money left in the budget, SimSE will assign a score to the budget.

4. Prerequisites

A student (a game player) should have a basic understanding of software engineering and management perspectives. (See Sections 6 and 7 of this document for ways to achieve this.)

For this game, participants do however, even though this game is called measurement game, but it is not necessary to have detailed knowledge about FSM measurement rules either from COSMIC or other FSM methods. The main objective of this game is to take management decisions on resources selection for requirements size measurement activities.

5. Time Commitment

The average time to play a single game (e.g., a waterfall SimSE) is 10-15 minutes, but, of course, it is likely to take several iterations (e.g. games) playing the game for the student to learn the concepts and be able to answer the questions. Participants should be given at least one week of out-of-class time to explore the game and answer the questions (see Section 8).

6. Suggested Supporting Lectures

The book 'Software Project Estimation' - by Alain Abran provides good insights for productivity model establishments.

7. Optional Supplementary Readings

1. Navarro, E.O. *"The Fundamental Rules" of Software Engineering.* 2008 [Available from: http://www.ics.uci.edu/~emilyo/SimSE/se_rules.html].

8. Assignment

Instructions

Step1) Watch the SimSE video tutorials at:

http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Videos.

Step 2) Download the SimSE player's manual at:

http://www.ics.uci.edu/~emilyo/SimSE/downloads.html#Docs.

Note 1: Be sure to watch the video and read the manual carefully, as they will highlight several important things that will significantly help you in successfully playing SimSE and correctly answering the questions.

Step 3) Next, download the Measurement game at:

https://chairegestionproduitsprojetslogiciels.github.io/simse-software-estimation-capability/games/game-b-early-sizing.html#downloads.

The download consists of a "readme" text file and an executable game, which you can run by simply double-clicking on it. If you do not have the current version of Java installed on your machine, you will have the opportunity to install it when you try to run a game.

Questions

We recommend choosing approximately three of the following questions.

1. What will be influenced based on the two different levels of FUR documentation?
2. How can I figure out whether the two measurers selected will finish the measurement activity within the budget?
3. Why is the average budget per requirement for the set of 50 requirements is lower than the average per requirement for the set of 10 requirements?
4. What is a measurer's salary based on?
5. What will be the measurement impact in term of measurement speed and measurement error when:
    a. one of the measurers selected has a perfectionist profile and a minor level expertise in measurement, and
    b. the other measurer has an expeditious profile but a major level of expertise in measurement?

9. How to Use This SimSE Module with other Modules

This module has been successfully used in conjunction with other measurement-related SimSE modules, making an assignment that consists collectively of three models/modules and associated questions

10. Other Notes

There are several other potentially effective uses for SimSE, most of which have yet to be fully explored:

- Have more advanced students modify an existing model (or build one from scratch, which should only be used with extremely advanced students) using SimSE's Model Builder tool and one of the existing models (available at http://www.ics.uci.edu/~emilyo/SimSE/downloads.html).

    - This has been tried, and results published in T. Birkhoelzer, E. Oh Navarro, and A. van der Hoek. *Teaching by Modeling instead of by Models*. Sixth International Workshop on Software Process Simulation and Modeling, May 2005 (available at http://www.ics.uci.edu/~emilyo/papers/ProSim2005.pdf).
- Our experience has suggested that an observer presence can have a positive effect on learning in SimSE. Although we have not tried this ourselves in classroom settings (only in controlled experiment settings), some suggested ways to try this are having students play SimSE in pairs, or having them play SimSE in a lab setting while observed by an instructor or TA.
- Have students play in teams, especially teams that have also done, or are doing, a class project together. This can add both a collaborative aspect to learning and, if set up to be a competition between teams, can add a competitive aspect.
- Make the assignment mandatory, rather than optional or extra-credit, to increase motivation.
- Have students play in a lab setting, both to add a competitive aspect and to allow them to collaborate.
    - Keep in mind, however, that a lab setting generally does not provide enough time to play a game enough to be able to answer all the questions.
    - An appropriate approach might be to allow students to play the game first in a lab session (this would also allow them to ask any questions that may arise), and then let them complete the rest of their playing and question-answering out of class.
- If a project is also being done as part of the course, have students pick one or more of the SimSE models and write an essay on comparisons between the SimSE process model(s) and the one followed in their project.

11. Feedback?

If you have any comments, suggestions, feedback, or experience regarding this course module or SimSE in general, please send an email to Alain Abran<Alain.Abran@etsmtl.ca>

References

1. COSMIC. (2020). Early Software Sizing with COSMIC: Practitioners Guide.

2. COSMIC. (2020). Early-Sizing-Experts-Guide-May-2020-1. https://doi.org/10.13140/RG.2.1.4195.0567

3. Abran, A. (2015). Software project estimation: The fundamentals for providing high quality information to decision makers. *John Wiley & Sons.*